

Human genetic ancestry, *Mycobacterium tuberculosis* diversity and tuberculosis disease severity in Dar es Salaam, Tanzania

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eLife Assessment

This **valuable** observational study was conducted in Dar es Salaam, Tanzania, to investigate potential associations between genetic variation in *Mycobacterium tuberculosis* and human host vs. disease severity. The authors conclude that human genetic ancestry did not contribute to tuberculosis severity and the evidence supporting this is generally **convincing**. The findings have significance for the understanding of the influence of host/bacillary genetics on tuberculosis disease.

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Abstract

Infectious diseases have affected humanity for millennia and are among the strongest selective forces. Tuberculosis (TB) is an ancient disease, caused by the human-adapted members of the *Mycobacterium tuberculosis* complex (MTBC). The outcome of TB infection and disease is highly variable, and co-evolution between human populations and MTBC strains may account for some of this variability. Particular human genetic ancestries have been associated with higher susceptibility to TB, but socio-demographic aspects of the disease can confound such associations. Here, we studied 1,000 TB patients from Dar es Salaam, Tanzania, together with their respective MTBC isolates, by combining human and bacterial genomics with clinical data. We found that the genetic background of the TB patient population was strongly influenced by migrations of Bantu-speaking populations from West Africa, which is in contrast to the corresponding MTBC genotypes that were mainly

introduced from outside Africa. These findings suggest a recent evolutionary history of co-existence between the human and MTBC populations in Dar es Salaam. We detected no evidence of an effect of human genetic ancestry, or MTBC phylogenetic diversity alone, nor their interaction, on TB disease severity. There was also no evidence of an association between human variation genome-wide and TB disease severity. Treatment-seeking, social and environmental factors are likely to be the main determinants of disease severity at the point of care in this patient population.

Author Summary

Tuberculosis (TB) is an ancient infectious disease that continues to challenge global health efforts. Here, we explored the interplay between human and bacterial genetics on TB in Dar es Salaam, Tanzania. We found that neither the genetic ancestry of the patient, nor the bacterial genotype alone, nor their interaction, influenced the severity of TB. Our finding indicate that in this patient population, social and environmental factors may be the main determinants of TB disease severity.

Introduction

Africa harbors the largest human genetic diversity worldwide (1). This continent is also inhabited by numerous ethnic and linguistic groups (2). While the long evolutionary history of modern humans in Africa and their large effective population sizes (3) explain this high genetic diversity, more recent migration events within and from outside Africa during the last 5,000 years, as well as admixture between historically separated populations, have resulted in some degree of homogenization (4). Hence, African populations nowadays are often composed of a mixture of different genetic ancestries (5, 6). One human migration that had a major influence on the population structure of present-day Africans is the so-called ‘Bantu expansion’, where Bantu-speaking groups migrated from central Western Africa southwards and eastwards, spreading farming technologies across sub-Saharan Africa, and admixing with local groups of hunter-gatherers and pastoralists (4, 7). Recent human genetic studies have identified moderate population structuring among Bantu-speaking populations (8, 9), yet, admixture with local populations has impacted immune responses to infectious diseases (10). Since pathogens have been one of the strongest selective forces driving human evolution (11), disease susceptibility and clinical outcomes can differ markedly between human populations. Genome-wide association studies (GWAS) have identified mutations altering the susceptibility to various infectious diseases (12–15), including tuberculosis (TB), which remains the main cause of human death due to a single infectious agent (16).

The bacteria that cause TB belong to the *Mycobacterium tuberculosis* complex (MTBC) and can be classified into ten human-adapted phylogenetic lineages: Lineage 1 (L1) to L10, plus several lineages adapted to different wild and domestic animal host species (17, 18). While TB is a global problem, the human-adapted MTBC lineages differ in their geographical distribution. L2 and L4 occur worldwide and other lineages are restricted to specific regions. Specifically, L1 and L3 mainly occur around the Indian Ocean (19), L5 and L6 that are limited to West Africa (20), and L7 only occurs in Ethiopia (21). This phylogeographic population structure of the human-adapted MTBC led to the hypothesis that certain MTBC genotypes are locally adapted to their sympatric human populations (19). This hypothesis is supported by findings from cosmopolitan settings, where sympatric associations between the geographical origin of TB patients and their infecting MTBC strains were observed (22–25). Moreover, in immune-compromised individuals with HIV co-infection, these sympatric associations were disrupted (26). The various

MTBC genotypes also differ phenotypically (27 [↗](#)), regarding disease progression (28 [↗](#)), transmission (29 [↗](#), 30 [↗](#)), disease presentation (31 [↗](#)) and immune activation in vitro (32 [↗](#), 33 [↗](#)) and ex vivo (34 [↗](#)).

Human genetic diversity has also been linked to differences in TB susceptibility. While for example *TYK2* has been associated with TB disease worldwide (35 [↗](#)), several human genetic loci were not consistently associated with TB in populations from different geographical regions but specific to certain populations (36 [↗](#)–40 [↗](#)). Particular human genetic ancestries have also been found to play a role in the context of TB. People with higher proportions of native Peruvian genetic ancestry showed a higher risk of progressing to active TB (41 [↗](#)), and a higher proportion of San genetic ancestry was associated with an increased risk for TB among South African Coloured individuals (42 [↗](#)). Differences in ethnicity have also been associated with different inflammatory profiles of TB patients at the time of presentation (34 [↗](#)). In addition to the effects of human and bacterial genetic diversity on TB, many social and environmental factors, as well as co-morbidities are known drivers of TB. These include malnutrition (43 [↗](#)), HIV infection, diabetes (44 [↗](#), 45 [↗](#)), poverty (46 [↗](#)), smoking (47 [↗](#)), and alcohol consumption (48 [↗](#)). While associations between TB and individual host, pathogen or environmental factors have been found (36 [↗](#)–41 [↗](#), 43 [↗](#)), studies considering all these components simultaneously, remain scarce (49 [↗](#)–51 [↗](#)).

Symptomatic TB has a wide spectrum of severity, and studies conducted before anti-TB drugs became available, suggest that mild disease is associated with higher odds of natural recovery (52 [↗](#), 53 [↗](#)). If mortality and natural recovery from symptomatic TB vary depending on the spectrum of disease, conceivably human populations exhibit genetic variation underlying disease severity. Furthermore, given that humans and the MTBC have coexisted for millennia, some of this variation could be linked to human ancestry. Here, we characterized the genetic variation including the genetic ancestry of a cohort of TB patients from Dar es Salaam, Tanzania, the phylogenetic lineage of their correspondent MTBC isolate, and investigated the association of both with TB disease severity while accounting for demographic, social-economical and environmental variables.

Results

Genetic ancestry of Tanzanian TB patients

Genetic ancestries were estimated for 7,479 individuals from 249 populations (Supplementary Figure S2) including 1,444 Tanzanian TB patients, using the software Admixture (54 [↗](#)) with 53,255 SNPs (Figure 1 [↗](#), Supplementary Figure S1).

The optimal number of source populations to describe our dataset was 24, based on the lowest cross-validation error (55 [↗](#)) (Supplementary Figure S3). For the purpose of this study, we named the genetic ancestries according to the geographical distribution and/or ethnicity of the reference populations that they are most prevalent in. The genetic ancestry with the highest contribution among Tanzanians with a mean of 44% (maximum 68%, minimum 0%) was also the most abundant in Bantu-speaking (BS) people from Southern and southeastern Africa (e.g. the Ronga population in Figure 1 [↗](#)), and hence hereafter, we will refer to this ancestry as “Southeastern BS” (Figure 1 [↗](#)). The second most common genetic ancestry with a mean of 22% (maximum 42%, minimum 6%) in the Tanzanian TB patients was most common among Kenyans (e.g. the Luhya population in Figure 1 [↗](#), 1000G and HGDP, see methods), and will be referred to as the “Eastern BS” genetic ancestry. Additionally, the Tanzanian TB patients had a mean of 9% (maximum 53%, minimum 0%) of a genetic ancestry that was most common among Bantu-speaking populations from western Central Africa, (e.g. the Eviya population in Figure 1 [↗](#)); we will refer to it as the “Western BS” genetic ancestry. Furthermore, the Tanzanian TB patients contained on average 4% of a genetic ancestry most abundant among Nigerians represented by the Esan population

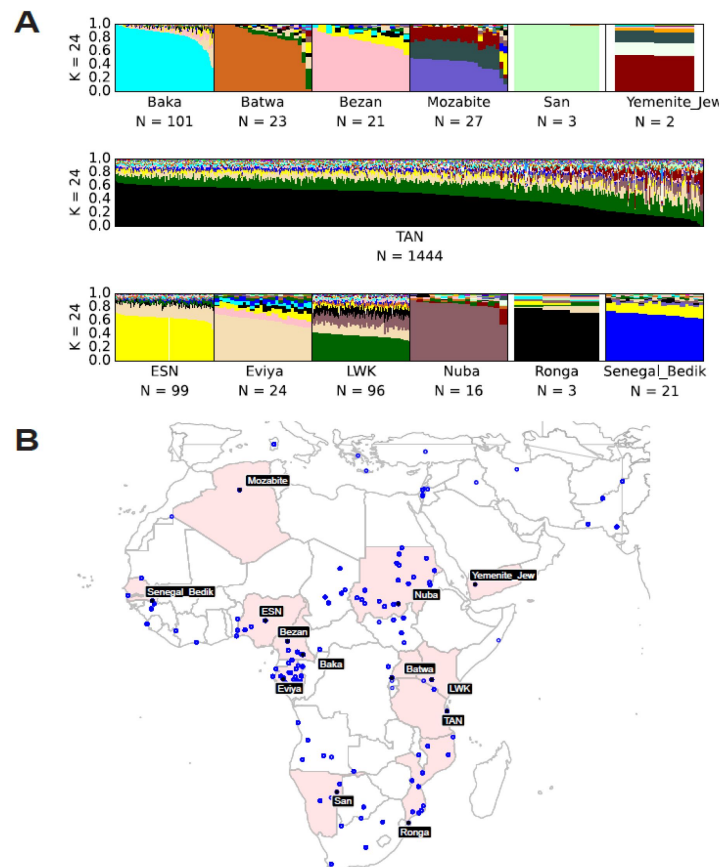


Figure 1

Genetic ancestry analyses of Tanzanian TB patients.

A) Genetic ancestry proportions of the 1,444 Tanzanian TB patients and representative human populations who shared at least 1% of their most common genetic ancestry with the Tanzanians for K = 24. (ESN: Esan from Nigeria (1000G), LWK: Luhya from Kenya (1000G)). For all populations included in our study, see Supplementary Figure S2 for their geographic distribution and Figure S5 for the ancestry composition of all African populations included in this study. **B)** The geographical location of the representative populations shown in A are depicted with black circles, and the corresponding country is highlighted. The remaining African populations included in the analysis are represented by blue circles.

(“Nigerian” genetic ancestry in [Figure 1](#)), and 4% of a genetic ancestry most abundant in people from Chad and Sudan (represented by the Nuba population in [Figure 1](#)). In addition, the genetic ancestry of the Tanzanian TB patients was composed of 3% of a genetic ancestry most prevalent in people from Western Africa (Gambia and Senegal represented by the Senegal Bedik population, in [Figure 1](#)), as well as 3% of a genetic ancestry most prevalent in individuals from western Africa rainforest hunter-gatherer populations (Bezan population in [Figure 1](#)). A mean of 2% belonged to a genetic ancestry most common among Bedouin individuals (represented by the Yemenite Jew population in [Figure 1](#)). The proportions of the remaining genetic ancestries were all smaller than 2% (Supplementary Figure S4, admixture plots for all African populations included can be found in Supplementary Figure S5). Finally, most Tanzanian TB patients had little non-African genetic ancestry (mean 5%, maximum 65%), with only 14 patients (~1%) showing more than 30% non-African genetic ancestry. In summary, the ancestry of Tanzanian TB patients was, for the most part, a mixture of three different Bantu components. Thus, for the remaining sections, we will focus on the ancestries termed Eastern BS (eBS), Southeastern BS (seBS), and Western BS (wBS).

Insights into the Bantu-speaking genetic ancestries in Africa and Tanzania

Compiling several datasets, including many different BS populations, allowed for a closer look at the distribution of BS ancestries across African populations. Like Tanzanians, the populations from the neighboring Kenya and Mozambique showed strong contributions of BS ancestries resulting from different admixture events ([Figure 1](#)). While the eBS genetic ancestry was highest in Kenya and Tanzania, decreasing from there to the south and to the west of the continent ([Figure 2A](#)), the seBS genetic ancestry generally increased towards the south and decreased towards the west as observed by others ([Figure 2B](#)) ([8](#)). The wBS genetic ancestry was mainly seen in BS populations from Gabon and Cameroon, as well as in South African populations ([Figure 2C](#)). At a continental-wide level, the geographical distribution and the genetic distances of the human populations analyzed were significantly correlated (Mantel test: veridical correlation = 0.18, p-value < 0.001, Supplementary Figure S6).

Even though all our TB patients were recruited in Dar es Salaam, we found them to belong to a variety of self-defined ethnic groups linked to different geographical regions within Tanzania ([Figure 2D](#)). Even at this smaller scale, we found a significant correlation between the geographic distance of the self-defined ethnic groups of our TB patients and their genetic distances (Mantel test: veridical correlation = 0.12, p-value < 0.001). The geographical structuring of human genetic diversity within Tanzania was further supported by our finding that the seBS genetic ancestry increased from West to East, and from North to South. The eBS genetic ancestry increased from South to North and decreased from West to East. The wBS genetic ancestry increased to the North and decreased to the East (Supplementary Figure S6A).

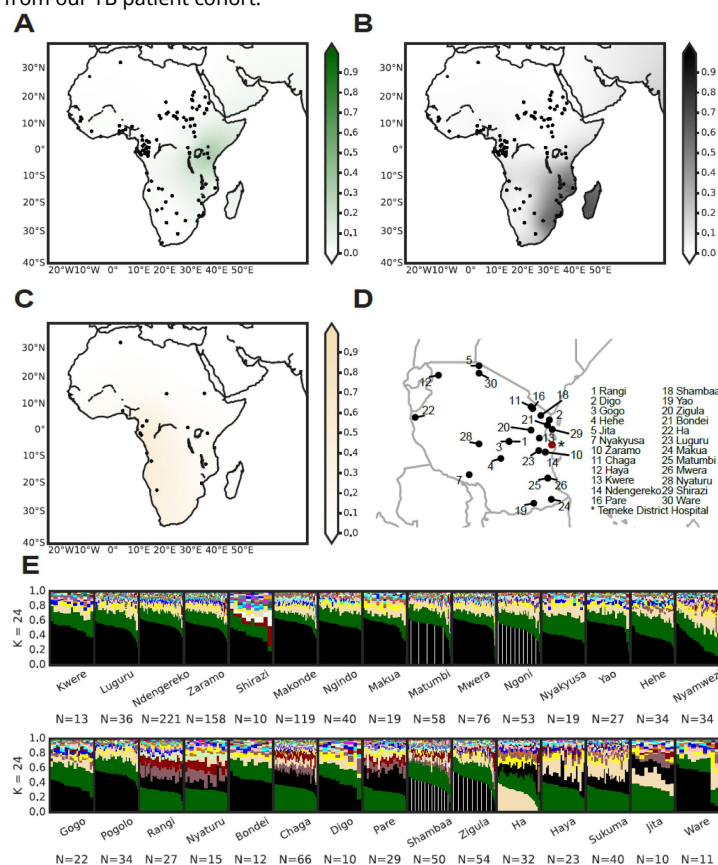
The MTBC genotypes circulating in Tanzania and their association with TB disease severity

In previous work ([56](#)), we investigated the MTBC genotypes causing TB in the patient cohort analyzed here. We found a high diversity of MTBC genotypes, with approximately half of the TB cases caused by four main MTBC genotypes estimated to have been introduced into Tanzania starting 320 years ago. After adding the genomic information of an additional 389 MTBC isolates to our dataset (new total N=1,471), the prevalence of the four dominant MTBC genotypes was very similar to our previous findings ([50](#)). The most frequent genotype within L3.1.1 (referred to as “Introduction 10”) contributed 39% of the current TB cases, followed by a genotype within L1.1.2 (“Introduction 9”) with 9%, a genotype within L4.3.4 (“Introduction 5”) with 6%, and a genotype within L2.2.1 (“Introduction 1”) with 5%. The remaining TB cases were caused by a variety of other genotypes within L1-L4, but occurred at frequencies of 1% or less ([56](#)). Despite a 36% increase in sample size compared to our previous analysis ([56](#)), we still found no evidence of an

Figure 2

Spatial visualizations of the BS genetic ancestries and the genetic ancestries of the different self-identified ethnic groups among the TB patients in Tanzania.

The genetic ancestry was inferred by admixture with $K = 24$, and the interpolation of the ancestries was performed by using the pykrige module in Python (see methods). **A)** eBS genetic ancestry, **B)** seBS genetic ancestry, and **C)** wBS genetic ancestry. The populations included for spatial interpolations are marked with a black dot on the maps. The maps were created using the basemap module in Python. **D)** Geographical origin of the ethnic groups among our TB patient cohort. The Temeke District hospital in Dar es Salaam where the patients were recruited is marked with a red point. Note that for some ethnic groups, no geographical origin could be identified (Supplementary Table 1). **E)** Ancestry plots for the different ethnic groups with at least 10 patients from our TB patient cohort.



association between the different MTBC genotypes circulating in Dar es Salaam and TB disease severity using as proxies; X-ray scores (mild or severe), bacterial load (inferred from GeneXpert cycle threshold) and TB-score (mild or severe) ([Table 1](#), [Table 3](#)) (see Methods).

Relationship between the human and MTBC population structures

We previously found that the four dominant MTBC genotypes in Dar es Salaam differed in their transmission rate and in the duration of the infectious period ([56](#)). Here, we assessed whether there might be an additional host genetic contribution to these differences. We first compared the genetic ancestry proportions between patients infected with the four dominant genotypes, and then tested whether patients who were genetically more closely related were infected with MTBC genotypes that were also more closely related as would be expected from a co-evolutionary process ([57](#)). However, the human genetic ancestry proportions differed only marginally between the TB patients infected by the four main MTBC genotypes ([Table 2](#)). Moreover, there was no correlation between the human and bacterial genetic distances (Mantel test: veridical correlation = -0.02, p-value = 0.85). Taken together, we found no statistically significant relationship between the human and bacterial genetic population structure in Dar es Salaam. These results also suggest that the genetic composition of this human population is unlikely to have a measurable effect on the differences in bacterial transmission rate and duration of the infectious period reported previously ([56](#)).

Association of human genetic ancestry with TB disease severity

In a previous publication on the same cohort, we found that the bacterial genetic background could not explain the differences observed in TB disease severity ([56](#)). Since TB disease is shaped by the bacterial and the human genetic background as well as environmental factors ([58](#)), we next investigated whether human genetic ancestry could have contributed to the differences in disease severity observed between our TB patients. We assessed the associations between ancestry and the three proxies of TB severity in HIV-negative patients using logistic regression models. We included the three human genetic ancestries with the highest proportions among the Tanzanian TB patients (seBS, eBS, wBS) as covariates, together with age, sex, smoking, socioeconomic status, TB category (relapse or not), malnutrition, education level, and drug resistance status to control for potential confounding. We found no evidence of an association between human genetic ancestry and any of these three proxies of TB disease severity ([Table 1](#)). We repeated the analysis by adding cough duration as covariate to possibly account for the disease duration before recruitment but the results remained unchanged. Additionally, we conducted a genome-wide association study to test for associations between specific human genetic variants and the three proxies of TB disease severity. We included the top three human genetic principal components (PCs), HIV status and the same covariates as for the regression described above. No evidence of an association was found (Supplementary Figure S7).

The combined effect of human and MTBC genetic diversity on TB disease severity

For a subset of 1,000 TB patients, we had both an MTBC genome and a human genome or genotype available. Genetic interactions between the host and the pathogen have been shown to affect TB severity in other settings ([41](#), [59](#)). To test for potential interactions between human and bacterial diversity on TB severity, we added to the ancestry logistic regression models described in the previous section the most common MTBC genotype as an additional explanatory variable (L3.1.1-Introduction 10), as well as the interaction between human ancestry and MTBC genotype. We only tested L3.1.1-Introduction 10, since the numbers of patients with the other genotypes were too few for meaningful testing. However, we found no evidence for any interaction between the MTBC genotype and human ancestry influencing TB disease severity in this patient population ([Table 3](#), Supplementary Figure S8).

Total N (%)	levels	Total N (%)	Missing N (%)	TB-score		Total N	Lung damage (X-ray score)				Bacterial load (Ct-value)		
				Severe N (%)	Mild N (%)		Missing N	severe	mild	not available	Total N	Missing N	Mean (SD)
MTBC Genotype	other	1471 (77.3)	433	207 (42)	406 (41)	764 (74.5)	262	51 (39)	269 (43)	293 (41)	863 (78.3)	239	19.4 (4.9)
	L2.2.1 - Intro 1			22 (4)	49 (5)			7 (5)	28 (4)	36 (5)			20.5 (5.1)
	L3.1.1 - Intro 10			184 (38)	388 (40)			59 (45)	239 (38)	274 (39)			19.1 (4.7)
	L4.3.4 - Intro 5			26 (5)	60 (6)			4 (3)	38 (6)	44 (6)			19.5 (4.1)
	L1.1.2 - Intro 9			51 (10)	78 (8)			11 (8)	58 (9)	60 (8)			19.1 (4.7)
				0.44				0.45	0.43				
seBS Bantu	Mean (SD)	1442 (75.7)	462	0.44 (0.2)	0.44 (0.2)	840 (81.9)	186	0.45 (0.1)	0.43 (0.2)	0.45 (0.2)	810 (73.5)	292	19.9 (5.2)
eBS	Mean (SD)	1442 (75.7)	462	0.23 (0.1)	0.22 (0.1)	840 (81.9)	186	0.22 (0.1)	0.23 (0.1)	0.22 (0.1)	810 (73.5)	292	19.9 (5.2)
wBS	Mean (SD)	1442 (75.7)	462	0.08 (0.1)	0.09 (0.1)	840 (81.9)	186	0.08 (0.1)	0.09 (0.1)	0.08 (0.1)	810 (73.5)	292	19.9 (5.2)
Other ancestry	Mean (SD)	1442 (75.7)	462	0.25 (0.1)	0.25 (0.1)	840 (81.9)	186	0.25 (0.1)	0.25 (0.1)	0.25 (0.1)	810 (73.5)	292	19.9 (5.2)

Table 1

Human and bacterial genotypes by the severity measures

Table 2

Characteristics of MTBC genotypes for all patients with either human or bacterial genetic data available.

	N (%) ¹	Missing N	levels	Other genotype	L2.2.1 - Intro 1	L3.1.1 - Intro 10	L4.3.4 - Intro 5	L1.1.2 - Intro 9	No bacterial data available
Total N (%)				613 (32)	71 (4)	572 (30)	86 (5)	129 (7)	433 (23)
Sex ²	1471 (100.0)	0	male (%)	424 (69)	52 (73)	425 (74)	55 (64)	87 (67)	302 (70)
			female (%)	189 (31)	19 (27)	147 (26)	31 (36)	42 (33)	131 (30)
Age in years	1471 (100.0)	0	Median (IQR)	33.0 (28.0-40.0)	31.0 (24.5-38.0)	33.0 (26.0-41.0)	31.5 (27.0-38.8)	35.0 (26.0-45.0)	35.0 (27.0-43.0)
HIV Status ²	1452 (98.7)	19	infected (%)	90 (15)	10 (14)	103 (18)	19 (22)	31 (25)	97 (23)
Smoker ²	1443 (98.1)	28	yes (%)	127 (21)	18 (26)	149 (27)	11 (13)	26 (20)	97 (23)
Cough duration (weeks)	1454 (98.8)	17	Median (IQR)	4.0 (3.0-4.0)	4.0 (3.0-4.0)	4.0 (3.0-4.0)	3.5 (2.2-4.0)	4.0 (2.0-5.0)	3.0 (2.0-4.0)
Socioeconomic status	1452 (98.7)	19	Median (IQR)	80.0 (50.0-125.0)	75.0 (50.0-125.0)	87.5 (50.0-126.2)	85.4 (50.0-122.9)	75.0 (50.0-125.0)	83.3 (50.0-133.3)
Education ²	1471 (100.0)	0	no education (%)	90 (14.7)	5 (7.0)	76 (13.3)	10 (11.6)	15 (11.6)	76 (17.6)
			primary (%)	391 (63.8)	54 (76.1)	390 (68.2)	61 (70.9)	92 (71.3)	273 (63.0)
			secondary (%)	105 (17.1)	12 (16.9)	90 (15.7)	13 (15.1)	18 (14.0)	64 (14.8)
			university (%)	27 (4.4)	0 (0.0)	16 (2.8)	2 (2.3)	4 (3.1)	20 (4.6)
Malnutrition ²	1471 (100.0)	0	yes (%)	83 (13.5)	9 (12.7)	71 (12.4)	11 (12.8)	13 (10.1)	56 (12.9)
Relapse/Reinfection ²	1447 (98.4)	24	yes (%)	9 (1.5)	4 (5.6)	18 (3.2)	2 (2.4)	3 (2.3)	12 (2.8)
Drug resistance status ²	1471 (100.0)	0	susceptible (%)	574 (93.6)	70 (98.6)	556 (97.2)	80 (93.0)	119 (92.2)	0
			INH-Mono (%)	37 (6.0)	1 (1.4)	16 (2.8)	6 (7.0)	10 (7.8)	0
			MDR (%)		0 (0.0)	0 (0.0)	0 (0.0)	2 (0.3)	0
seBS	1009 (68.6)	462	Median (IQR)	0.5 (0.3-0.6)	0.5 (0.4-0.5)	0.5 (0.4-0.6)	0.5 (0.4-0.6)	0.5 (0.4-0.6)	0.5 (0.3-0.6)
eBS	1009 (68.6)	462	Median (IQR)	0.2 (0.2-0.3)	0.2 (0.2-0.3)	0.2 (0.2-0.3)	0.2 (0.2-0.3)	0.2 (0.2-0.3)	0.2 (0.2-0.3)
wBS	1009 (68.6)	462	Median (IQR)	0.1 (0.0-0.1)	0.1 (0.1-0.1)	0.1 (0.0-0.1)	0.1 (0.0-0.1)	0.1 (0.0-0.1)	0.1 (0.0-0.1)
Other ancestry	1009 (68.6)	462	Median (IQR)	0.2 (0.2-0.3)	0.2 (0.2-0.3)	0.2 (0.2-0.3)	0.2 (0.2-0.2)	0.2 (0.2-0.2)	0.2 (0.2-0.3)

¹ The column N (%) indicates the total number of patients with bacterial genetic data available that contained a value for the respective variable.

² The percentage of a MTBC genotype that have the respective characteristic (e.g percentage of males among patients infected with an Intro 1 MTBC genotype).

		Disease severity measure					
		Lung damage		TB-score		Bacterial load (Ct-value)	
		OR adjusted	p-value adjusted	OR adjusted	p-value adjusted	OR adjusted	p-value adjusted
MTBC genotype ¹	L3.1.1-Introduction	1.60 (0.95-2.67)	0.07	1.00 (0.72-1.40)	0.98	0.99 (0.97-1.00)	0.13
	other genotypes	1		1		1.00	
Human ancestry ²	seBS category 3	2.29 (1.04-5.42)	0.19	1.06 (0.66-1.69)	0.17	1.00 (0.98-1.03)	0.13
	seBS category 2	2.40 (1.15-5.41)		0.62 (0.40-0.94)		1.02 (0.99-1.04)	
	seBS category 1	1		1		1.00	
	eBS category 3	1.30 (0.51-3.79)		1.09 (0.64-1.84)		1.02 (0.99-1.05)	
	eBS category 2	1.50 (0.61-4.28)		1.17 (0.69-1.94)		1.00 (0.97-1.03)	

Table 3

Estimated associations between disease severity, human genetic ancestry and bacterial genotype.

Three variables as proxies for disease severity were included: Lung damage (mild versus severe), TB-score (mild versus severe), Bacterial load (continuous, log10 transformed). Binomial logistic regressions were performed on the data of HIV-negative patients and adjusting was done for age, sex, smoking, socioeconomic status, level of education, malnutrition, TB Type (relapse or new infection), and drug resistance status by including these variables in the model. For the ancestries and the interactions, the p-values were retrieved by performing a likelihood ratio test comparing a model including the ancestries and interactions to a model without them. This table combines the results of two logistic regressions per disease severity measure, one including an interaction and one without. The ancestries were transformed and categorized (see Methods) with category 1 comprising the lowest amount of the respective ancestry and category 3 (4 in the case of wBS) the highest amount.

Table 3 (continued)

	eBS category 1	1		1		1.00	
	wBS category 4	0.58 (0.23-1.53)		0.94 (0.49-1.76)		1.02 (0.98-1.06)	
	wBS category 3	0.54 (0.21-1.46)		1.09 (0.57-2.04)		1.03 (0.99-1.07)	
	wBS category 2	0.73 (0.26-2.15)		1.03 (0.50-2.09)		1.03 (0.99-1.08)	
	wBS category 1	1		1		1.00	
Interaction ²	seBS category 3* L3.1.1-Intro 10	1.05 (0.20-5.43)	0.06 ⁽¹⁾	1.05 (0.40-2.71)	0.92	0.99 (0.94-1.04)	0.83
	seBS category 2* L3.1.1-Intro 10	0.39 (0.08-1.94)		1.29 (0.52-3.15)		0.98 (0.93-1.03)	
	seBS category 1* L3.1.1-Intro 10	1		1		1.00	
	eBS category 3* L3.1.1-Intro 10	8.32 (0.91-193.14)		1.45 (0.48-4.34)		1.01 (0.94-1.07)	
	eBS category 2* L3.1.1-Intro 10	6.38 (0.71-146.86)		1.25 (0.42-3.70)		1.02 (0.96-1.09)	
	eBS category 1* L3.1.1-Intro 10	1		1		1.00	
	wBS category 4* L3.1.1-Intro 10	0.17 (0.02-1.23)		1.79 (0.49-6.56)		0.97 (0.90-1.05)	
	wBS category 3* L3.1.1-Intro 10	0.33 (0.04-2.51)		1.48 (0.40-5.51)		0.99 (0.92-1.07)	
	wBS category 2* L3.1.1-Intro 10	0.87 (0.09-8.06)		2.40 (0.53-11.05)		0.98 (0.90-1.07)	
	wBS category 1* L3.1.1-Intro 10	1		1		1.00	

¹ The odds ratio for genotype represents the odds of severe disease for Introduction 10 compared to the odds for other genotypes.

² The odds ratios represent the estimated multiple in the odds of severe disease for a one-unit increase in the additive log-transformed ancestry variable.

Discussion

In this study, we analyzed the genetic ancestry of TB patients, the MTBC diversity underlying their TB infection, and estimated the associations of both with disease severity in Dar es Salaam, Tanzania. We found a strong component of BS genetic ancestries among the Tanzanian TB patients, similar to those of neighboring populations from Mozambique and Kenya, and little non-African genetic ancestry. Genetic ancestry proportions did not differ between patients infected with different MTBC genotypes. There was no evidence that the patient genetic ancestry or the MTBC genotype on their own, nor their interaction, had any effect on TB disease severity.

Despite the fact that Tanzania is one of the few countries in sub-Saharan Africa where all four main African linguistic groups co-exist (60), and that its largest city and economic capital, Dar es Salaam, is strongly influenced by different human populations from within and outside Africa, our cohort of TB patients mostly comprised BS ethnicities. Comparing this TB patient cohort to a large number of modern human populations revealed major components of eBS and seBS genetic ancestries. This genetic population structure probably resulted from several admixture events estimated to have happened between 1500 to 150 years ago, between local populations and BS populations who migrated from West Africa to the East and South of the continent (4). The TB patients investigated here were recruited in one district hospital of Dar es Salaam. Yet, we found the genetic distances between the patients to be correlated with the original geographic range of their self-identified ethnicities, suggesting that the corresponding human populations are not fully admixed. The population of Dar es Salaam has increased by several millions in the last 40 years, mainly as a result of immigration from rural parts of Tanzania (61). Thus, our findings suggest that our TB patient population mostly represents recent migrants to Dar es Salaam from other regions of Tanzania. Moreover, we found little evidence of Eurasian genomic influence in the TB patient population (on average 5% genetic ancestry). This is in strong contrast to coastal Swahili populations, as recent findings comparing modern and medieval Swahili people revealed large components of genetic ancestry derived from exchanges between local East African BS populations and people from India, Persia, and Arabia, starting as early as 1000 AD (62). We conclude that our TB patient population does not represent the full spectrum of human genetic diversity in Tanzania.

In contrast to the genetic ancestry of the TB patients, we found that the MTBC genotypes infecting these patients descend from multiple historical introductions, which mainly resulted from the human exchanges that took place across the Indian Ocean during the last few centuries (56). Some of these recently introduced MTBC genotypes became dominant, in particular the MTBC genotype L3.1.1-Introduction 10, which caused TB in approximately 40% of our patients. These strains descended from an introduction that occurred approximately 300 years ago from South or Central Asia to East-Africa (56). Hence, while our TB patient population reflects little historical gene flow from non-African populations, the underlying MTBC diversity indicates that the MTBC genotypes introduced from outside successfully spread in this newly encountered host population, eventually outcompeting native MTBC genotypes (63).

We previously reported that in this TB patient cohort, some of the dominant MTBC genotypes had a higher transmission rate than others, while some other MTBC genotypes induced patients to remain infectious for longer (56). Based on the similar proportions of MTBC genotypes among self-reported ethnic groups we observed at the time, we had already hypothesized that human genetic heterogeneity of the host population is unlikely to be responsible for those differences (56). Here we formally addressed this hypothesis, and found that there was no evidence that the human genetic ancestry proportions differed between patients infected with different MTBC

genotypes in our cohort. This finding is consistent with the notion that the differences in epidemiological parameters we reported previously are mainly determined by the pathogen genotype.

Disease severity is one aspect of the clinical presentation of TB with a direct impact on patient mortality and morbidity, as well as on pathogen transmission, as it influences patient infectiousness. It is thus likely that both host and pathogen genetic characteristics can modulate TB disease severity (57 [↗](#)). Experimental infections in various animal models suggest that different MTBC strains vary in virulence (64 [↗](#)–66 [↗](#)). However, *in clinico* evidence for differences in disease severity caused by different MTBC genotypes are inconsistent (67 [↗](#), 68 [↗](#)). We found no evidence of differences in disease severity at the point of care caused by the different MTBC genotypes in our study. Moreover, we did not observe any association between human genetic ancestry and disease severity, which is in contrast to a recent study from Peru, where human genetic ancestry was found to influence progression to active TB (41 [↗](#)).

Our previous work found evidence of such genetic interactions when considering the complete genomes of the TB patients and their infecting MTBC strain (50 [↗](#)), and identified associations between human and pathogen variants. Such associations reflect host-pathogen genetic interactions that determine susceptibility to symptomatic TB or intra-host selection during mycobacterial replication. Here, in the context of co-evolutionary history between humans and MTBC, we specifically tested whether an interaction between the main human ancestry components and being infected with the most dominant MTBC strain in Dar es Salaam could explain the variability in TB disease severity. However, we did not find evidence of such an effect. Others have reported an association between TB disease severity, a particular bacterial genotype and a particular human SNP in Uganda, but did not explicitly link this human SNP to a particular human genetic ancestry (59 [↗](#), 69 [↗](#)). Several factors could account for the lack of effect we observed. First, our patient population was relatively genetically homogeneous, given that the different BS components represent populations with only moderate levels of genetic differentiation (70 [↗](#)). Second, there is likely to be selection bias in our cohort since only patients presenting at the clinic were recruited. The disease severity measures included in this study mainly reflect disease stages at which patients felt ill enough to go to the hospital; i.e. we did not consider intermediate, more contrasting disease states that are known to occur between infection and the development of symptoms (71 [↗](#)). To at least partially account for that, we included the number of weeks a patient was coughing as a covariate in our analyses. Third, the lack of a measurable interaction between host genetic ancestry and MTBC genotype could reflect the relatively recent presence of these MTBC genotypes in Tanzania, and the distinct (i.e. allopatric) geographical origins of the host and pathogen populations. This indicates that none of the ancestral human populations that compose modern Tanzanians has lived in sympatry with the ancestors of the modern MTBC genotypes that circulate in Dar es Salaam today.. With the exception of West Africa, where the geographically restricted West-African MTBC lineages L5 and L6 remain an important cause of human TB (72 [↗](#)), the situation in Tanzania might be representative of the TB epidemics in many African countries, as evidence suggests that many MTBC genotypes dominating the continent today have been introduced from outside Africa recently in the history of human populations (63 [↗](#), 73 [↗](#)–75 [↗](#)).

In conclusion, our study shows that the TB patients from Dar es Salaam were mainly of BS genetic ancestry reflecting limited Eurasian genetic influx. Neither the human genetic ancestry or the MTBC genotype alone, nor their interaction, were associated with TB disease severity. Our results highlight the dominant role of social and environmental factors in human TB in Tanzania.

Methods

Study population

This study is based on a previously described dataset (50 [DOI](#), 56 [DOI](#)). Briefly, adult active TB patients with pulmonary disease (sputum smear-positive and GeneXpert-positive) were recruited between November 2013 and June 2022 at the Temeke District Hospital in Dar es Salaam, Tanzania when they first presented for care. Sputum and blood samples were collected from each patient to extract DNA for sequencing of the MTBC strain, and genotyping or whole-genome sequencing (WGS) of the patient. Additionally, clinical and sociodemographic information was obtained from every patient. In total, there was either human or bacterial data available for 1,904 patients. Of those 1,444 patients had human genetic data and 1,471 had bacterial genetic data available, respectively. A total of 1,000 patients had both types of data available after quality-based filtering. The geographical locations of the self-indicated ethnic group of each patient were retrieved by searching for the original region of the respective ethnic group, and if they originated from a single region, the geographic coordinates according to Wikipedia were taken. If two neighboring regions were among the origins, then a random location between the two regions was taken as surrogate (Supplementary Table S1).

Bacterial and human sample processing

The MTBC bacteria were cultured on solid Löwenstein-Jensen media at the TB laboratory of the Ifakara Health Institute in Bagamoyo, Tanzania. Before 2018, the MTBC isolates were transferred to Switzerland for DNA extraction and later, DNA extraction was carried out in Bagamoyo. Bacterial WGS was done using the Illumina short-read technology at the Department of Biosystems Science and Engineering of ETH Zurich in Basel (DBSSE). Human WGS was done at the Health 2030 Genome Center in Geneva, Switzerland using an Illumina NovaSeq 6000 sequencer. The human genotyping was done at the iGE3 Genomics platform at the University of Geneva in Switzerland using the Illumina Infinium H3Africa genotyping microarray (Version 2; <https://chipinfo.h3abionet.org> [DOI](#)) plus custom Tanzanian-specific SNP add-ons (76 [DOI](#)).

Human genetic data

The processing of the human genetic data has been described in detail by Xu et al (50 [DOI](#)). Briefly, we used the GRCh38 as a human reference genome to map the WGS reads of 118 patients using BWA aligner (v0.7.17) (77 [DOI](#)). Duplicate reads were then marked with the markduplicates module of Picard (v2.8.14) (78 [DOI](#)). Variant calling was first done for each sample individually following GATK best practices for germline short variant discovery. Samples with a coverage below 5 were excluded followed by a joint calling of the variants. A Quality Score Recalibration (VQSR) based filter was applied (real sensitivity of 99.7, excess heterozygosity of 54.69) and samples with more than 50% missing genotype calls were removed.

For the genotyping data, we used the Illumina GenomeStudio software (v2.0.5, https://support.illumina.com/array/array_software/genomestudio/downloads.html [DOI](#)) to analyze the raw microarray data. Samples with a low quality, that were badly clustered, or that had a call rate below 0.97, were excluded. The PLINK format was then converted to VCF format using PLINK (v1.9) (79 [DOI](#)). The first round of imputation was performed with the African Genome Resources (AFGR, <https://www.apcdr.org/> [DOI](#)) reference panel on the sanger imputation server with EAGLE (80 [DOI](#)) for phasing and Positional Burrows-Wheeler transform (PBWT) (81 [DOI](#)) for imputing. The second round was performed with a reference panel created in house that was based on 118 patients with available whole-genome sequences (76 [DOI](#)) with SHAPEIT4 for phasing and Minimac3 for imputing. For each SNP, the reference panel with the highest imputation quality score was used to determine the final genotype call. SNPs with an INFO score below 0.8 were discarded.

Bcftools (v1.15) was used to merge the WGS and genotyping samples after identifying SNPs shared between the two methods that were missing in fewer than 10 samples.

Whole-genome sequence analysis of the MTBC bacteria

We analyzed all FASTQ files using the WGS analysis pipeline described previously (82). In summary, Trimmomatic (83) v. 0.33 (SLIDINGWINDOW:5:20) was used to remove the Illumina adaptors and to trim low quality reads. Only reads with a length of at least 20 bp were kept for further analysis. Overlapping paired-end reads were merged using SeqPrep v. 1.2 (84) (overlap size = 15). We then mapped the resulting reads to a reconstructed ancestral sequence of the MTBC (85) with BWA v. 0.7.13 (mem algorithm) (77). Picard v. 2.9.1 (86) was then applied to mark and exclude duplicated reads. Furthermore, the RealignerTargetCreator and IndelRealigner modules of GATK v. 3.4.0 (87) were used to perform local realignment of reads around INDELs. Reads having an alignment score lower than $(0.93 \times \text{read length}) - (\text{read length} \times 4 \times 0.07)$, corresponding to more than 7 miss-matches per 100bp, were excluded using Pysam v. 0.9.0 (88). SNP calling was then performed with SAMtools v. 1.2 mpileup (89) and VarScan v. 2.4.1 (90) with the following thresholds: a minimum mapping quality of 20, a minimum base quality at a position of 20, minimum read depth at a position of 7 and no strand bias. Positions in repetitive regions such as PE, PPE, and PGRS genes or phages were excluded, as described in (91). A whole-genome Fasta file was created from the resulting VCF file. We applied some additional filters; genomes were excluded from downstream analysis if they had a sequencing coverage of lower than 15 or if they contained SNPs suggestive of different MTBC lineages (i.e. mixed infections). We identified lineages and sublineages using the SNP-based classification by Steiner et al. (92), and Coll et al. (93), respectively. In addition, we identified drug resistance mutations published in the WHO catalogue (94) and determined the respective drug resistance profile. The majority of strains were susceptible (95%), while 70 (5%) were isoniazid mono-resistant and two contained isoniazid and rifampicin resistance and were considered as multi-drug resistant.

Identification of bacterial SNPs diagnostic for the successful MTBC introductions

We previously identified several successful MTBC introductions into Dar es Salaam (49). For these, we aimed to obtain a set of diagnostic SNPs that would allow assigning MTBC strains not included in our previous study to these genotypes. For that, we merged the VCF files from the 1,082 MTBC genomes included in that previous dataset by using BCFtools (v1.9). We then used VCFtools (v0.1.16) to remove Indels and positions that were variable in less than 12 genomes (12 was the minimal threshold selected when identifying the successful introductions in our previous publication (56)). By using the R package VariantAnnotation (95) and a customized Python script, SNPs specific to one of the most successful introductions were extracted. To ensure the SNPs identified as markers for the introductions were specific, we also identified phylogenetic SNPs on a bigger and global dataset representing the human-adapted MTBC diversity (82), and tested whether any of the phylogenetic SNPs identified for any of the successful Introductions was present in any of the MTBC lineages or sublineages. We compiled a subset of 25 SNPs (Supplementary Table S2) that we used as phylogenetic markers for the different MTBC introductions and identified strains belonging to one of the four most successful MTBC introductions (Introduction 1, Introduction 5, Introduction 9, and Introduction 10) in the expanded MTBC dataset based on these SNPs by using a customized script.

Measures of TB disease severity

We used three different proxies for TB disease severity. The first one was the TB-score, which is a clinical score adapted from Wejse et al. (96) that consists of several signs and symptoms including the presence of fever and the Body Mass Index (BMI). A point was given for each of the following symptoms or clinical measures if present: BMI below 18, BMI below 16, mid upper arm circumference (MUAC) below 220, MUAC below 200, body temperature higher than 37°C, cough,

hemoptysis, dyspnea, chest pain, night sweat, abnormal auscultation, anaemia. A maximum of 12 points could be achieved, and a TB-score below 6 was considered as mild and everything above as severe. As a second proxy, we assessed the amount of lung damage. Two independent radiologists assessed chest X-ray pictures of the patients and gave a Ralph score (97 [4](#)). X-ray scores above 71 were considered as severe and everything below as mild. The Ralph score has been validated to grade chest X-ray severity in adult pulmonary TB patients and 71 was the optimal cut-off point for predicting unfavorable outcome (89 [4](#)). Furthermore Ralph scores higher than 71 have been associated with a longer duration of symptoms, a lower BMI, and higher clinical scores (99 [4](#)). As a third proxy we used the bacterial load in the sputum represented by the difference between the first (early cycle) and the last (late cycle) threshold during quantitative PCR (Ct-value) as determined by GeneXpert MTB/RIF assays. For each sputum sample, we took five probes, run a quantitative PCR each and reported the lowest Ct-value.

Genetic ancestry analysis of TB patients

To investigate the genetic ancestry of the TB patients, we combined our dataset with the data from ten other projects: The Gambian Genome Variation Project (GGVP) (100 [4](#)), the 1000 Genomes Project (1000G) (101 [4](#)), the Human Genome Diversity Project (HGDP) (102 [4](#)), Simons Genome Diversity Project (103 [4](#)), as well as data generated by Patin et al. 2014 and 2017 (4 [4](#), 104 [4](#)), Hollfelder et al. 2017 (105 [4](#)), Semo et al. 2019 (8 [4](#)), Schlebusch et al. 2012 (106 [4](#)), and Fortes-Lima et al. 2022 (107 [4](#)). We used the GRCh37 version of all datasets. The dataset of HGDP was in GRCh38 version and we thus did a lift over to GRCh37 using the picard (v2.26.10) (78 [4](#)) tool LiftoverVcf. For all the datasets including only populations from one single continent, we excluded variants with a missingness of more than 10% and only included variants that did not deviate from Hardy-Weinberg equilibrium ($p < 1e-5$) using PLINK (version 1.9b, www.cog-genomics.org/plink/1.9/) (79 [4](#)). For the 1000G, SGDP, and HGDP data, we first identified variants that deviated from the Hardy-Weinberg equilibrium ($p < 1e-5$) in each superpopulation using PLINK (version 2.0a, www.cog-genomics.org/plink/2.0/) and removed them from the whole dataset. We additionally removed variants with a high missingness ($> 10\%$) from the full datasets using PLINK (version 1.9b). After extraction of 103,262 nucleotide positions common to all datasets, we merged the datasets using PLINK (version 1.9b). From the merged dataset, we removed second-degree relatives using PLINK (version 2.0a, king cutoff of 0.088) (N=369) and patients from our cohort where the sex according to the genetic data did not correspond with the sex indicated in the clinical data, patients who were genetic outliers based on principal component analysis (PCA) or who did not cluster with any other African samples (N=83). In addition, we removed regions of high linkage disequilibrium ([https://genome.sph.umich.edu/wiki/Regions_of_high_linkage_disequilibrium_\(LD\)](https://genome.sph.umich.edu/wiki/Regions_of_high_linkage_disequilibrium_(LD))) and applied additional filters to the merged dataset (missingness $> 10\%$, minimum allele frequency of 5%, removal of sex chromosomes, variant pruning with `-indep-pairwise 50 10 0.1`, only biallelic positions) ending up with 53,255 positions and 7,479 individuals from 249 populations.

To infer the ancestry proportions of the Tanzanian TB patients, we used ADMIXTURE (version 1.3.0) (54 [4](#)). We estimated the number of ancestral populations (K) by running ADMIXTURE 15 times for each value for K from K=2 until K=29 with the option `--cv`. The `--cv` option performs 5-fold cross-validation and allows to identify the value for K resulting in the lowest cross-validation error (Supplementary Figure S3). The cross-validation error was lowest for K = 24. From the 15 runs performed with K = 24, we selected a representative of an output that was supported by most of the 15 runs (6/15) to extract the ancestry proportions of each individual. A PCA was performed using PLINK (version 1.9b) on all African populations included.

Spatial interpolation of human genetic ancestry proportions

To visualize the distributions of the different patient ancestries, we performed spatial interpolation using the OrdinaryKriging function from the pykrige module in Python (variogram_model = "linear", grid space of 500). For the eBS and seBS ancestries, we included all

African populations for the interpolation. For the wBS, the interpolation failed when using all African populations with an insufficient slope, suggesting little spatial variability. Since many populations were sampled in the region with the highest proportions of wBS genetic ancestry and among them many hunter-gatherer populations containing little to no wBS genetic ancestry, we repeated the interpolation with only a subset of the non-hunter-gatherer populations.

Correlation between distance matrices

To assess, whether MTBC genotypes that are more closely related tend to infect people that are also more closely related genetically which would be compatible with co-evolution, we investigated the correlation between the human and bacterial distance matrices. To calculate the pairwise bacterial genetic distances, alignments of variable positions where data was missing in less than 10% of the genomes were generated and used to create SNP distance matrices according to the Hamming distance (<https://git.scicore.unibas.ch/TBRU/tacos> ).

Insertions and deletions were considered as missing data. To get the human pairwise genetic distances, we calculated the Euclidean distance based on the first two principal components. When only looking at the Tanzanian TB patients, we calculated the principal components for the Tanzanians only while for the continental dataset we included all available African populations.

To investigate, whether human populations that are geographically more distant are also genetically more distant, we calculated the correlation between the geographical and the genetic distance on an African level as well as on the level of Tanzania. For the geographical distance matrix, we calculated the Euclidean distance based on the latitude and the longitude. At the level of Tanzania, the broad geographic location of the original area of the ethnic group was considered (Supplementary Table S1) and for the continental level, the coordinates of the hospital in Temeke were taken for the Tanzanian TB patients considering that for the other studies only the sampling locations were known. The human and bacterial genetic distance matrices as well as the human genetic and geographic distance matrices were tested for correlations using the `mantel.test()` function from the `mantel` module in Python (options: `perms = 10000`, `method = "pearson"`, `tail = "upper"`).

Statistical analyses

Clinical and sociodemographic characteristics, MTBC genotype as well as human genetic ancestry were summarized by the different disease severity measures using proportions or means and standard deviations. Similarly, the human genetic ancestries were summarized by MTBC genotype.

To estimate the associations between disease severity and the explanatory variables bacterial genotype (binary, belonging to Introduction 10 or not) and human ancestries (seBS, eBS, wBS), we used a logistic regression model. We included three variables as proxies for disease severity: Lung damage based on X-ray score (mild versus severe), TB-score (mild versus severe), bacterial load (continuous, log10 transformed). Categorization or transformation of the disease severity measures were performed because the distributions of the residues were not normal, violating the assumption of linear regression. We tested only for Introduction 10 because there were few observations of the other MTBC genotypes. To account for the compositional nature of the human ancestries (i.e. that they sum up to 1), we used the additive log transformation from the R package ‘`compositions`’ ([108](https://doi.org/10.1093/bioinformatics/bty108) ). The ancestry proportions were transformed and categorized with category 1 comprising the lowest amount of the respective ancestry and category 3 (4 in the case of wBS) the highest. The categories were chosen to have roughly equal numbers of patients in each. We used categories to allow a non-linear relationship without specifying polynomials, and avoid having difficulty in interpretation but we recognize that a small amount of information is lost. Similar results were also obtained with other parameterizations. We assessed whether there was an interaction between ancestry and genotype on TB severity using the likelihood ratio test. For that, we compared a model including the interaction between the MTBC genotype and human

genetic ancestry to a model without the interaction using the ‘lmtest’ package (109 [link](#)). The estimates were adjusted for: age, sex, smoking, the number of weeks with cough, the socioeconomic status (ratio between household income and number of household members), education (no, primary, secondary, university), drug resistance profile, TB category (new infection or relapse/reinfection), and malnutrition. Only HIV-negative patients were included in the analysis. All statistical analyses were carried out in R (version 4.1.2). Code for the statistical analysis can be found (https://github.com/mzwyrer/TB-Dar_Mtb [link](#)).

To conduct the genome-wide association study (GWAS), we used logistic regression (TB Score, X-ray Score) and a linear model (Ct-value) implemented by PLINK (version 1.9). A total of 6,665,541 common (MAF > 0.05) human genetic variants were included. Genetic principal components were calculated using GCTA (version 1.93.2). As covariates, we included the top three human genetic principal components (PCs) to correct for population stratification, along with sex, HIV status, age, cough duration, smoking, socioeconomic status, TB category, malnutrition, education, and drug resistance profile. Genomic inflation factors were below 1 of all tested outcomes (Supplementary Figure S9).

Data availability

The bacterial WGS data can be found under the bioprojects PRJEB49562 and PRJNA670836 on ENA and the human WGS and genotyping data under EGAS00001005850 and EGAS00001007216, respectively. Clinical data are available on request.

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Additional information

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Author contributions

MZ: Conceptualization, Data Curation, Formal Analysis, Project Administration, c, Visualization, Writing – Original Draft Preparation; ZMX: Resources, Writing – Review & Editing; AR: Formal Analysis, Methodology, Writing – Review & Editing; JH: Resources; MS: Resources; MR: Methodology, Writing – Review & Editing; HH: Resources, Writing – Review & Editing; LKR: Resources; MR: Methodology; SB: Resources; KR: Funding Acquisition, Writing – Review & Editing; JF: Funding Acquisition, Writing – Review & Editing; DP: Funding Acquisition, Writing – Review & Editing; LQM: Conceptualization, Writing – Review & Editing; SG: Funding Acquisition, Supervision, Project administration, Writing – Original Draft Preparation; DB: Supervision, Conceptualization, Funding Acquisition, Investigation, Methodology, Writing – Original Draft Preparation

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Additional files

Supplementary Files. Supplementary Figure S1 - The first two PCA components for all African populations included in this study ($n=116$). **Supplementary Figure S2** - Populations included in the Admixture analysis. Populations are coloured according to the study from which the data was obtained (4, 8, 44, 91–93, 95–98). **Supplementary Figure S3** - Boxplots of cross-validation errors for values of K between 2 and 29 resulting from 15 runs. The lowest cross-validation error was obtained for $K = 24$. **Supplementary Figure S4** - Boxplots of the proportions of the 24 genetic ancestries among the Tanzanian TB patients. The ancestries were named after the population(s) where they were most abundant. **Supplementary Figure S5** - Ancestry plots for all African populations included ($K = 24$). **Supplementary Figure S6** - Heatmap showing the correlations between the genetic ancestries and geographical location. A) Correlations at the country level of Tanzania between the latitude and longitude of the ethnic group the patients belong to. B) Correlations on a continental level including all African populations. For the latitude, a negative correlation indicates higher genetic ancestry proportions towards the South and for the longitude a negative correlation indicates lower genetic ancestry proportions towards the East. **Supplementary Figure S7** - Manhattan plot for GWAS conducted using A, B, C) TB-score, X-ray score, and Ct-value. Red line indicates GWAS significance threshold of $5e-8$. **Supplementary Figure S8** - Association between the Bantu genetic ancestries and TB-score and X-ray score for each of the most successful introductions. A, C, E) Southeastern, Eastern, and Western Bantu on mild or severe TB-score, B, D, E) Southeastern, Eastern, and Western Bantu genetic ancestry lung damage assessed by lung X-ray. **Supplementary Figure S9** - QQ Plot and genomic inflation factor for GWAS conducted using A, B, C) TB-score, X-ray score, and Ct-value. **Supplementary Table S1** - The different ethnic groups with at least 10 members in our cohort and the region and broad geographic location of the original area of the ethnic group. The latitude and longitude are given in decimal degrees. In the case of two associated regions, a location close to the border of the two regions was selected. **Supplementary Table S2** - Phylogenetic markers selected to identify the Introductions. The position is based on the reconstructed reference of the ancestor (79) and the derived base indicates the base present in the respective Introduction. Intro1 refers to Introduction 1 within L2.2.1, Intro5 to Introduction 5 within L4.3.4, Intro9 to Introduction 9 within L1.1.2, and Intro10 to Introduction 10 within L3.1.1. 

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Reviewer #2 (Public review):

Summary:

This manuscript reports the results of an observational study conducted in Dar es Salaam, Tanzania, investigating potential associations between genetic variation in *M. tuberculosis* and human host vs. disease severity. The headline finding is that no such associations were found, either for host / bacillary genetics as main effects or for interactions between them.

Strengths:

Strengths of the study include its large size and rigorous approaches to classification of genetic diversity for host and bacillus.

Comments on revisions:

The authors have responded satisfactorily to comments raised.

<https://doi.org/10.7554/eLife.103533.2.sa2>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

This Tanzanian study focused on the relationship between human genetic ancestry, Mycobacterium tuberculosis complex (MTBC) diversity, and tuberculosis (TB) disease severity. The authors analyzed the genetic ancestry of 1,444 TB patients and genotyped the corresponding MTBC strains isolated from the same individuals. They found that the study participants predominantly possess Bantu-speaking genetic ancestry, with minimal European and Asian ancestry. The MTBC strains identified were diverse and largely resulted from introductions from South or Central Asia. Unfortunately, no associations were identified between human genetic ancestry, the MTBC strains, or TB severity. The authors suggest that social and environmental factors are more likely to contribute to TB severity in this setting.

Strengths:

In comparison to other studies investigating the role of human genetics in TB phenotypes, this study is relatively large, with more than 1,400 participants.

The matched human-MTBC strain collection is valuable and offers the opportunity to address questions about human-bacterium co-evolution.

Weaknesses:

Although the authors had genome-wide genotyping and whole genome sequencing data, they only compared the associations between human ancestry and MTBC strains. Given the large sample size, they had the opportunity to conduct a genome-wide association study similar to that of Muller et al. (<https://doi.org/10.1016/j.ygeno.2021.04.024>).

Thank you very much for taking the time to carefully review our manuscript and for your suggestions and comments. In another published study using the same cohort (<https://doi.org/10.1101/2023.05.11.23289848>), we performed a genome-wide association analysis between the genome-wide SNPs of the host and the genome-wide SNPs from the paired MTBC strains. In the current work we were interested in testing specifically if host ancestry and pathogen genotype family, as well as their interaction, were associated with differences in disease severity, a clinical phenotype with direct consequences for both host and pathogen fitness. The study of Müller et al, referred to by the reviewer, investigates whether MTBC families of strains causing disease in two patient cohorts (South Africa and Ghana) were associated with particular human SNPs assessed genome-wide. In that study, clinical phenotypes were not assessed and human ancestries, in a much broader sense than the ones used in our current study, were used as covariates. To leverage the genome-wide information and the clinical variables collected in our study, we have now added a genome-wide association analysis of all the human SNPs with disease severity measures while adjusting for co-variables (age, sex, smoking, cough duration, socioeconomic status, history of previous TB, malnutrition, education level, and drug resistance status) and for human population stratification. Yet, no significant statistical associations were detected (L243-249).

The authors tested whether human genetic ancestry is associated with TB severity. However, the basis for this hypothesis is unclear. The studies cited as examples all focused on progression to active TB (from a latent infection state), which should not be conflated with disease severity. It is difficult to ascertain whether the role of genetic ancestry in disease severity would be detectable through this study design, as some participants might simply have been sicker for longer before being diagnosed (despite the inquiry about cough duration). This delay in diagnosis would not be influenced solely by human genetics, which is the conclusion of the study.

Evidence that mortality and natural recovery from TB vary by disease presentation spectrum come from studies carried out before the introduction of anti-TB chemotherapy. Patients with mild disease presentation, as measured by radiology at the time of diagnosis had higher odds of recovering naturally compared to those with advanced disease (doi: 10.5588/ijtld.23.0254, doi: 10.1164/arrd.1960.81.6.839). Given the deleterious effects of an MTBC infection leading to symptomatic disease on human fitness, we hypothesized that natural selection has acted on human traits underlying TB disease severity. If those traits are heritable one would expect to find underlying genetic variation in human populations. In addition, because certain MTBC genotype families and human populations have co-existed since at least a few centuries to a few millennia, we hypothesized that some of that genetic variation could be related to human ancestry. We have added more details to the introduction to make our rationale clearer (L118-127). In our patient cohort, we observed a large variation in disease severity using as approximations; TB-Score, X-Ray score and bacterial burden in sputa (Ct-value as determined with GeneXpert). However, the reviewer is absolutely correct in that patients in our study are being diagnosed at different stages of disease confounding our analysis. This is a limitation of our study which cannot be fully accounted for by including cough duration, as we also acknowledged in the manuscript (L343-346).

Additionally, the study only included participants who attended the TB clinic.

Yes, this is related to the previous point, our study only considers patients that felt ill enough to visit the TB clinic potentially not including patients that had less severe disease as acknowledged.

Including healthy controls from the general population would have provided an interesting comparison to see if ancestry proportions differ.

We agree that it would be interesting to compare the ancestries of healthy controls to the ancestries of TB patients from the same population. However, that would be especially informative with respect to TB susceptibility and would not necessarily be informing disease severity traits and its underlying genetics. The similarities between the ancestry proportions of our cohort with those of neighboring countries such as Kenya, Malawi and Mozambique publicly available genomic data, suggests that there would be no major differences between TB patients and healthy controls.

Although the authors suggest that social and environmental factors contribute to TB severity, only age, smoking, and HIV status were characterised in the study.

Based on the comments of both reviewers, we added the following additional variables as covariates in the regression models: the socioeconomic status representing the ratio between the household income and the number of individuals in the household, malnutrition, the education level and whether it was a relapse/reinfection or a new case.

Reviewer #2 (Public review):

Summary:

*This manuscript reports the results of an observational study conducted in Dar es Salaam, Tanzania, investigating potential associations between genetic variation in *M. tuberculosis* and human host vs. disease severity. The headline finding is that no such associations were found, either for host / bacillary genetics as main effects or for interactions between them.*

Strengths:

Strengths of the study include its large size and rigorous approaches to classification of genetic diversity for host and bacillus.

Weaknesses:

(1) There are some limitations of the disease severity read-outs employed: X-ray scores and Xpert cycle thresholds from sputum analysis can only take account of pulmonary disease. CXR is an insensitive approach to assessing 'lung damage', especially when converted to a binary measure. What was the basis for selection of Ralph score of 71 to dichotomise patients? If outcome measures were analysed as continuous variables, would this have been more sensitive in capturing associations of interest?

Thank you very much for taking the time to carefully review our manuscript and for your suggestions and comments.

We recruited active TB patients with pulmonary TB disease that were sputum smear-positive and GeneXpert-positive. In this study we aimed at obtaining paired samples from both the patient and the strain, and in the current analysis we aimed at testing if human ancestry and

its interaction with the strain genotype could explain differences in disease severity. It is often difficult to obtain microbiological cultures from extra-pulmonary cases and including those cases would have not been possible at the scale of this cohort. We believe as well that extra-pulmonary TB is of less relevance for the question we are addressing because in exclusively extrapulmonary cases, disease severity is not linked with bacterial transmission. However, extra-pulmonary TB can be extremely severe, and it would be very interesting to explore the potential role of human genetic variation underlying extra-pulmonary TB in future studies.

As to the insensitivity of CXR to measure lung damage, we would argue that it depends on what is being assessed. As a rationale for the Ralph score, its inventors argue that as in other grading methods, the proportion of affected lung and or cavitation is important to assess severity. It has been described as a “validated method for grading CXR severity in adults with smear-positive pulmonary TB that correlates with baseline clinical and microbiological severity and response to treatment, and is suitable for use in clinical trials” (<https://thorax.bmj.com/content/thoraxjnl/65/10/863.full.pdf>). While the validation of the score is convincing in that study, and the score has been used in several TB studies and trials, the low proportion of HIV co-infections might have been a limitation. Indeed, as shown in our previous publication, in our cohort of patients, chest X-ray scores were significantly lower in HIV infected TB patients <https://doi.org/10.1371/journal.ppat.1010893>. In the current analysis, regression analyses performed for the CXR severity and for the other severity measures did not include HIV co-infected patients.

We obtained the same pattern of results using a continuous outcome. However, an assumption of linear regression was violated. The residuals were not normally distributed stemming from the bimodal distribution of the scores in our dataset. The threshold of 71 for the Ralph score has been used by others in previous studies; in its original description it has been suggested as the optimal cut-off point for predicting a positive sputum smear status after two months, which in turn has been shown to predict unfavorable outcomes (<https://doi.org/10.1136/thx.2010.136242>). Another study showed that a Ralph score higher than 71 was significantly associated with a longer duration of symptoms, higher clinical scores and a lower BMI (doi: 10.5603/ARM.2018.0032).

(2) There is quite a lot of missing data, especially for TB scores - could this have introduced bias? This issue should be mentioned in the discussion.

While we have a TB-score available for each patient, the chest X-ray score is missing for many patients. However, this is random and due both to the absence of an X-ray picture or to the bad quality of X-ray pictures that the radiologists could not assess. When stating that there is a lot of missing data for the TB scores, we assume that the reviewer was referring to the “missing N” columns in Table 1. There, the number of observations missing in each of the disease severity measures actually relates to the explanatory variables (i.e MTBC genotype and human ancestries). This table includes all patients that either had a bacterial genome available or a human genome/genotype (N = 1904). As an example for the TB-score as outcome variable, for 1471 patients the MTBC genotype was determined while it was missing for 433 patients. On the other hand for X-ray scores, 177 had a severe X-ray score, 849 a mild one and for 878 patients, there was no X-ray score available. As for the Ct-value, despite the fact that the patients were recruited based on positive GeneXpert by the clinical team, these results were not always available to us.

(3) The analysis adjusted for age, sex, HIV status, age, smoking and cough duration - but not for socio-economic status. This will likely be a major determinant of disease severity. Was adjustment made for previous TB (i.e. new vs repeat episode) and drug-sensitivity of the isolate? Cough duration will effectively be a correlate/consequence of more severe disease - thus likely highly collinear with disease severity read-outs - not a true

confounder. How does removal of this variable from the model affect results? Data on socioeconomic status should be added to models, or if not possible then lack of such data should be noted as a limitation.

Out of the 1904 patients that have either human or bacterial genomic data available, 48 were relapses (2.5%). The mean of the disease severity measures suggest that relapses have a higher CXR score but the TB-score and Ct-values did not differ. Based on the comments of both reviewers, we added the following additional variables as covariates to the regression models: the socioeconomic status representing the ratio between the household income and the number of individuals in the household, malnutrition examined by a doctor, the education level, and whether it was a relapse/reinfection or a new case and if the causative strain had any resistance to any anti-TB drugs. The results did not change. Cough duration could also be a consequence of more severe disease, as pointed out by the reviewer. We present now the results excluding cough duration as a variable from the model, however this also did not affect the results.

(4) Recruitment at hospitals may have led to selection bias due to exclusion of less severe, community cases. The authors already acknowledge this limitation in the Discussion however.

(5) Introduction: References refer to disease susceptibility, but the authors should also consider the influences of host/pathogen genetics on host response - both in vitro (PMIDs 11237411, 15322056) and in vivo (PMID 23853590). The last of these studies encompassed a broader range of ethnic variation than the current study, and showed associations between host ancestry and immune response - null results from the current study may reflect the relative genetic homogeneity of the population studied.

We thank the reviewer for these suggestions which we have added to the introduction.

Reviewer #1 (Recommendations for the authors):

Minor Comments:

(1) The authors should be careful when using the term "Bantu" as opposed to "Bantu-speaking". (i.e. referring to the language group). The term is considered offensive in some settings.

We thanks the reviewer for this important concern, we have revised throughout the manuscript.

(2) There are several "(Error! Reference source not found)" phrases in the place of references throughout the document.

We thank the reviewer for pointing this out, this has been corrected in the revised version.

(3) Please correct line 365: "... sequencing (WGS) the patient...." to "... sequencing (WGS) of the patient...."

(4) The figures in the supplementary PDF are not numbered and some are cut-off (I think it is Supplementary Figure S2).

This has been corrected in the revised version.

Reviewer #2 (Recommendations for the authors):

Typographical errors

| (1) *There are multiple instances where references have not pulled through to the text, e.g. line 126 (Error! Reference source not found.)*

We thank the reviewer for pointing this out, this has been corrected in the revised version.

| (2) *Line 239: have been show - have been shown?*

Thank you, this mistake has been corrected in the revised version.

<https://doi.org/10.7554/eLife.103533.2.sa0>